MILITARY SPECIFICATION

TRANSISTOR, PNP, SILICON, DUAL

TYPE 3N93

1. SCOPE

1.1 Description.- This specification covers the detail requirements for a silicon, PNP, dual-emitter transistor, characterized for low-level, chopper applications, and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Physical dimensions.- See figure 1 (TO-72).

1.3 Maximum ratings.-

P_T 1/ T_A = 25°C	T_{stg}	V_{E_1C} or V_{E_2C}	V_{E_1B} or V_{E_2B}	$V_{E_1E_2}$	V_{CB}
mW	°C	Vdc	Vdc	Vdc	Vdc
300	-65 to +200	-50	-50	±50	-50

I_C	I_B	I_{E_1} or I_{E_2}
mAdc	mAdc	mAdc
±20	±20	±10

1/ Derate linearly at 1.71 mW/°C above T_A = 25°C.

1.4 Primary electrical characteristics.-

	$\|V_o\|$	$r_{e_1e_2}$(on)	h_{fe}
	I_B = -1.0 mAdc	I_B = -1.0 mAdc	V_{CE_1} (or V_{CE_2}) = -6.0 Vdc
	$I_{E_1} = I_{E_2} = 0$ (See figure 2)	$I_{E_1} = I_{E_2} = 0$ I_{e_1} = 100 μAac f = 1 kc	I_C = -1.0 mAdc I_{E_2} (or I_{E_1}) = 0 f = 4 Mc
	μVdc	ohm	----
Min.	----	10	1.5
Max.	50	50	----

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 5961

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SPECIFICATION

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification For.

STANDARDS

MILITARY
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions.- The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500 and as follows:

$BV_{E_1E_2}$	Breakdown voltage, emitter one to emitter two, collector short-circuited to base.
BV_{ECO}	Breakdown voltage, emitter to collector, base open.
I_e	Emitter current (r.m.s.).
$I_{E_1E_2}$	Emitter to emitter cutoff current, collector short-circuited to base.
$r_{e_1e_2}$ (on)	Dynamic series impedance between emitter one and emitter two.
$V_{E_1E_2}$	Emitter one to emitter two voltage (d.c.).
$\lvert V_O \rvert$	Magnitude of emitter to emitter offset voltage (d.c.).
$\lvert \Delta V_O \rvert_{I_B}$	Magnitude of algebraic difference between offset voltages measured at two different base currents.
$\lvert \Delta V_O \rvert_{T_A}$	Magnitude of algebraic difference between offset voltages measured at two different temperatures.
Offset voltage	The emitter to emitter offset voltage is defined as the open circuit voltage appearing between the two emitters when the collector-base junction is forward biased.

3.3 Design, construction and physical dimensions.- Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics.- Performance characteristics shall be as specified in 4.3 and 4.4.

3.5 Marking.- The following marking specified in MIL-S-19500 may be omitted from the body of the transistor:

a. Manufacturer's identification.
b. Country of origin.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.3 and 4.4. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.1).)

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4.2 Quality conformance inspection.- Quality conformance inspection shall consist of the examinations and tests specified in 4.3 and 4.4.

4.3 Group A inspection.- Group A inspection shall consist of the examinations and tests specified in table I.

4.4 Group B inspection.- Group B inspection shall consist of the examinations and tests specified in table II.

4.5 Methods of examination and test.- Methods of examination and test shall be as specified in tables I and II, and as follows:

4.5.1 Open-circuit input capacitance.- Method of test shall be in accordance with Method 3240 of MIL-STD-750 except that the output capacitor shall be omitted, and the alternate emitter shall be open circuited.

4.5.2 Emitter to emitter cutoff current.- The transistor shall be tested in the circuit shown on figure 4. The voltage specified herein shall be applied between the two emitter terminals. The current following into emitter one shall then be measured.

4.5.3 Emitter to collector breakdown voltage.- Method of test shall be in accordance with method 3011 of MIL-STD-750, Condition D except that the collector and the specified emitter terminal shall be interchanged and the alternate emitter shall be open circuited.

4.5.4 Emitter to emitter breakdown voltage.- The transistor shall be tested in the circuit shown on figure 4. The resistor R_1 is a current limiting resistor and should be of sufficiently high resistance to avoid excessive current flowing through the transistor and the current meter. The voltage shall be gradually increased from zero until the minimum limit of $BV_{E_1E_2}$ or the test current specified herein is reached. The transistor is acceptable if the minimum limit for $BV_{E_1E_2}$ is reached before the test current reaches the value specified herein. If the test current specified herein is reached first, the device shall be rejected.

4.5.5 Emitter to emitter offset voltage.- The transistor shall be tested in the circuit shown on figure 2. The base current shall be adjusted to the value specified herein. The voltage between the two emitters shall then be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

4.5.6 Emitter to emitter offset voltage change with base current.- The offset voltage and its polarity shall be measured as specified in 4.4.5 at the two base currents specified herein. The magnitude of the algebraic difference between these two values shall then be computed.

4.5.7 Emitter to emitter offset voltage change with temperature.- The offset voltage and its polarity shall be measured as specified in 4.5.5 at the two temperatures specified herein. The magnitude of the algebraic difference between these two values shall then be computed.

4.5.8 Dynamic "on" series impedance.- The transistor shall be tested in the circuit shown on figure 3. The base current shall be adjusted to the value specified herein and an a.c. sinusoidal signal current of the rms value specified herein applied through both emitters. The rms voltage, $V_{e_1e_2}$, between the two emitters shall be measured using an a.c. voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The dynamic "on" series impedance shall then be determined as follows:

$$r_{e_1e_2(on)} = \frac{V_{e_1e_2}}{I_{e_1}}$$

where $V_{e_1e_2}$ is the rms voltage between the two emitters.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

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6. NOTES

6.1 The activity responsible for the qualified products lists is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.)

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - SH
(Project 5961-N039(NAVY))

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Table I - Group A inspection.

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Subgroup 1			10				
Visual and mechanical examination	2071						
Subgroup 2			5				
Emitter to base cutoff current	3061 Cond. D	V_{E_1B} = -25 Vdc I_C = 0 I_{E_2} = 0		I_{E_1BO}	---	-0.1	nAdc
Emitter to base cutoff current	3061 Cond. D	V_{E_2B} = -25 Vdc I_C = 0 I_{E_1} = 0		I_{E_2BO}	---	-0.1	nAdc
Emitter to emitter cutoff current	(See 4.5.2)	$V_{E_1E_2}$ = +25 Vdc V_{CB} = 0		$I_{E_1E_2}$	---	+0.1	nAdc
Emitter to emitter cutoff current	(See 4.5.2)	$V_{E_1E_2}$ = -25 Vdc V_{CB} = 0		$I_{E_1E_2}$	---	-0.1	nAdc
Collector to base breakdown voltage	3001 Cond. D	I_C = -1.0 μAdc $I_{E_1} = I_{E_2}$ = 0		BV_{CBO}	-50	---	Vdc
Emitter to collector breakdown voltage	(See 4.5.3)	I_{E_1} = -1.0 μAdc I_{E_2} = 0 I_B = 0		BV_{E_1CO}	-50	---	Vdc

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Table I - Group A inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Emitter to collector breakdown voltage	(See 4.5.3)	$I_{E_2} = -1.0\ \mu Adc$ $I_{E_1} = 0$ $I_B = 0$		BV_{E_2CO}	-50	---	Vdc
Emitter to base breakdown voltage	3026 Cond. D	$I_{E_1} = -1.0\ \mu Adc$ $I_{E_2} = 0$ $I_C = 0$		BV_{E_1BO}	-50	---	Vdc
Emitter to base breakdown voltage	3026 Cond. D	$I_{E_2} = -1.0\ \mu Adc$ $I_{E_1} = 0$ $I_C = 0$		BV_{E_2BO}	-50	---	Vdc
Emitter to emitter breakdown voltage	(See 4.5.4)	$I_{E_1} = +1.0\ \mu Adc$ $V_{CB} = 0$		$BV_{E_1E_2}$	+50	---	Vdc
Emitter to emitter breakdown voltage	(See 4.5.4)	$I_{E_1} = -1.0\ \mu Adc$ $V_{CB} = 0$		$BV_{E_1E_2}$	-50	---	Vdc
Collector to base cutoff current	3036	$V_{CB} = -25$ Vdc $I_{E_1} = I_{E_2} = 0$		I_{CBO}	---	-10.0	nAdc
Subgroup 3			5				
Magnitude of emitter to emitter offset voltage	(See 4.5.5)	$I_B = -1.0$ mAdc $I_E = I_{E_2} = 0$ $T_A = 25°C$		$\lvert V_o \rvert$	---	50	Vdc

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Table I - Group A inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Magnitude of emitter to emitter offset voltage	(See 4.5.5)	I_B = -1.0 mAdc $I_{E_1} = I_{E_2} = 0$ T_A = -25°C		$\lvert V_o \rvert$	---	50	µVdc
Magnitude of emitter to emitter offset voltage	(See 4.5.5)	I_B = -1.0 mAdc $I_{E_1} = I_{E_2} = 0$ T_A = +100°C		$\lvert V_o \rvert$	---	50	µVdc
Magnitude of emitter to emitter offset voltage change with base current	(See 4.5.6)	I_{B_1} = -1.5 mAdc I_{B_2} = -0.5 mAdc $I_E = I_{E_2} = 0$		$\lvert \Delta V_o \rvert$	---	25	µVdc
Magnitude of emitter to emitter offset voltage change with temperature	(See 4.5.7)	I_{B_1} = -1.0 mAdc $I_{E_1} = I_{E_2} = 0$ $T_A(1)$ = 100°C $T_A(2)$ = -25°C		$\lvert \Delta V_o \rvert$	---	75	µVdc
Dynamic "on" series impedance	(See 4.5.8)	I_{B_1} = -1.0 mAdc $I_{E_1} = I_{E_2} = 0$ I_{e_1} = 100 µAac rms f = 1 kc		$r_{e_1e_2}(on)$	10	50	ohms
Subgroup 4			10				
Small-signal short-circuit forward current transfer ratio	3306	V_{CE_1} = -6.0 Vdc I_C = -1.0 mAdc		h_{fe_1}	1.5	---	---

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Table I - Group A inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Small-signal short-circuit forward current transfer ratio (Cont'd)		$I_{E_2} = 0$ f = 4 Mc					
Small-signal short-circuit forward current transfer ratio	3306	$V_{CE_2} = -6.0$ Vdc $I_C = -1.0$ mAdc $I_{E_1} = 0$ f = 4 Mc		h_{fe_2}	1.5	---	---
Open-circuit output capacitance	3236	$V_{CB} = -6.0$ Vdc $I_{F_1} = I_{E_2} = 0$ f = 4 Mc		C_{obo}	---	10	pf
Open-circuit input capacitance	3240 (See 4.5.1)	$V_{E_1B} = -6.0$ Vdc $I_C = 0$ $I_{E_2} = 0$ f = 4 Mc		C_{ibo_1}	---	3.0	pf
Open-circuit input capacitance	3240 (See 4.5.1)	$V_{E_2B} = -6.0$ Vdc $I_C = 0$ $I_{E_2} = 0$ f = 4 Mc		C_{ibo_2}	---	3.0	pf
Emitter to emitter cutoff current	(See 4.5.2)	$V_{E_1E_2} = +25$ Vdc $V_{CB} = 0$ $T_A = 100°C$		$I_{E_1E_2}$	---	+10	nAdc

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Table I - Group A inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Emitter to emitter cutoff current	(See 4. 5. 2)	$V_{E_1E_2}$ = -25 Vdc V_{CB} = 0 T_A = 100° C		$I_{E_1E_2}$	---	-10	nAdc
Subgroup 5			10				
Delay time		(See figure 5)		td	---	20	ns
Rise time		(See figure 5)		tr	---	50	ns
Storage time		(See figure 5)		ts	---	250	ns
Fall time		(See figure 5)		tf	---	75	ns

Table II - Group B inspection.

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
Subgroup 1			20				
Physical dimensions	2066	See figure 1					
Subgroup 2			15				
Solderability	2026	Omit aging					
Thermal shock (temperature cycling)	1051 Cond. C						
Thermal shock (glass strain)	1056 Cond. A						
Moisture resistance	1021						
Subgroup 3			15				
Shock	2016	1500 G, 0. 5 msec 5 blows each orientation: X_1, Y_1, Y_2, Z_1					

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Table II - Group B inspection (Cont'd).

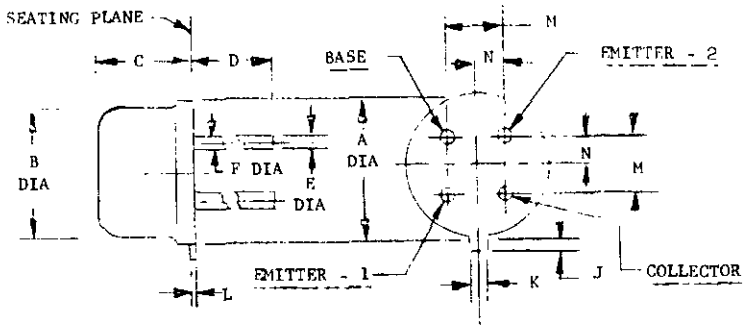
EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Vibration fatigue	2046	nonoperating					
Vibration, variable frequency	2056						
Constant acceleration (centrifuge)	2006	20,000 G each orientation: X_1, Y_1, Y_2, Z_1					
End points for subgroups 2 and 3:							
Magnitude of emitter to emitter offset voltage	(See 4.5.5)	I_{B_1} = -1.0 mAdc $I_{E_1} = I_{E_2} = 0$		$\lvert V_o \rvert$	---	50	μVdc
Emitter to base cutoff current		$V_{E_1E_2}$ = +25 Vdc $V_{CB} = 0$		$I_{E_1E_2}$	---	+0.5	nAdc
Emitter to base cutoff current		$V_{E_1E_2}$ = -25 Vdc $V_{CB} = 0$		$I_{E_1E_2}$	---	-0.5	nAdc
Subgroup 4			10				
Terminal strength (lead fatigue)	2036 Cond. E						
End point:							
Seal	MIL-STD-202 Method 112	Test cond. C Procedure III Test cond. A for gross leaks		leak rate	---	$5x10^{-7}$	atm cc per sec
Subgroup 5			15				
Salt atmosphere (corrosion)	1041						
Subgroup 6			λ = 10				
High-temperature life (nonoperating)	1031	T_A = 200°C					

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Table II - Group B inspection (Cont'd).

EXAMINATION OR TEST	MIL-STD-750 METHOD	MIL-STD-750 DETAILS	LTPD	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
Subgroup 7			$\lambda = 10$				
Steady-state operation life	1026	$V_{CE} = -10$ Vdc $V_{E_1E_2} = 0$ $P_T = 300$ mW $T_A = 25°C$					
End points for subgroups 6 and 7:							
Magnitude of emitter to emitter offset voltage	(See 4.5.5)	$I_{B_1} = -1.0$ mAdc $I_{E_1} = I_{E_2} = 0$		$\lvert V_o \rvert$	---	60	μVdc
Emitter to base cutoff current		$V_{E_1E_2} = +25$ Vdc $V_{CB} = 0$		$I_{E_1E_2}$	---	+1.0	nAdc
Emitter to base cutoff current		$V_{E_1E_2} = -25$ Vdc $V_{CB} = 0$		$I_{E_1E_2}$	---	-1.0	nAdc
Dynamic "on" series impedance	(See 4.5.8)	$I_{B_1} = -1.0$ mAdc $I_{E_1} = I_{E_2} = 0$ $I_{e_1} = 100$ μAac rms f = 1 kc		$r_{e_1e_2}$(on)	---	60	ohms

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DIMENSIONS					NOTES
LTR	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.209	.230	5.31	5.84	
B	.178	.195	4.52	4.95	
C	.170	.210	4.32	5.33	
D	.500	----	12.70	----	7
E	----	.021	----	.53	2, 7
F	.016	.019	.41	.48	3, 7
J	.028	.048	.71	1.22	6
K	.036	.046	.91	1.17	
L	----	.020	----	.51	
M	.0707	Nom	1.80	Nom	4
N	.0354	Nom	0.90	Nom	4

NOTES:

1. Metric equivalents in table and parentheses are shown for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250(6.35) from the seating plane.
3. Measured in the zone .050(1.27) and .250(6.35) from the seating plane.
4. When measured in a gaging plane $.054^{+.001}_{-.000}(1.37^{+.03}_{-.00})$ below the seating plane of the transistor, max dia leads shall be within .007(.18) of their true location relative to a maximum width tab. Smaller dia leads shall fall within the outline of the max dia lead tolerance.
5. The collector shall be internally connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All 4 leads.

Figure 1 - Physical dimensions of transistor type 3N93 (TO-72).

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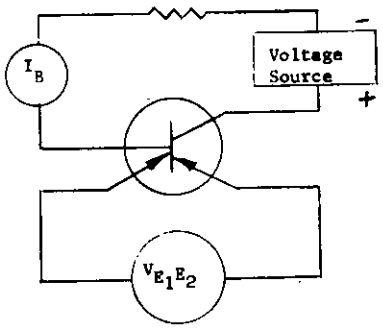


Figure 2 - Offset voltage test.

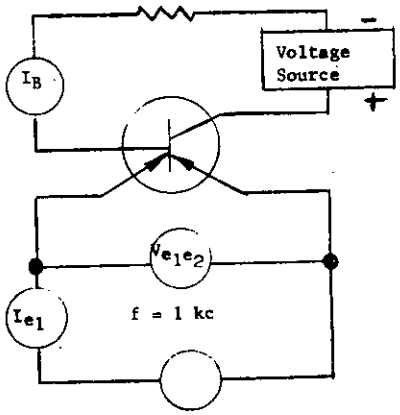


Figure 3 - Dynamic "on" series impedance test.

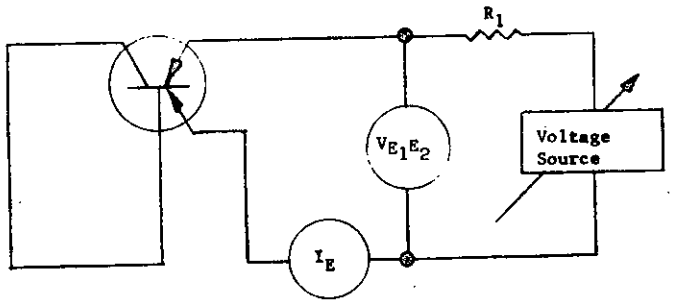
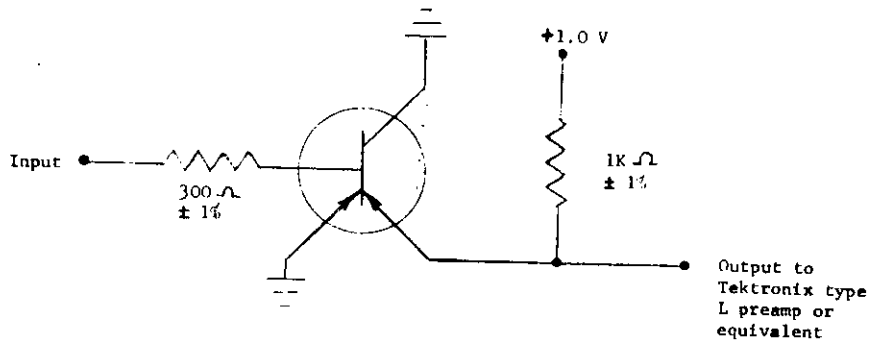


Figure 4 - Emitter to emitter cutoff current and Emitter to emitter breakdown voltage test.

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Input Pulse = + 1 volt
PW ≥ 1 μsec
t_r ≤ 10 ns
t_f ≤ 10 ns
rep. rate = 1 kc

Figure 5 - Switching time test circuit.

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